AVR PROGRAMMER USING AVR910 PROTOCOL



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Newbies often have difficulty getting started with programming of AVR microcontroller memory. What they need is a simple documentation and the entire code for AVR memory programming.

This article attempts to simplify the existing AVR910 technical note available on ATMEL's website. It includes a source program for the microcontroller written in 'C' language for easy understanding of the principles and a user interface software program written in Microsoft Visual Basic .Net for interfacing and transferring a binary code to the target device.

The complete system for programming an AVR microcontroller uses a universal asynchronous receiver transmitter (UART) with serial peripheral interface (SPI) to program the target microcontroller's fuses, lock bits, flash or EEPROM. Any AVR board can be converted into a master programmer using this protocol.

Circuit description

Fig. 1 shows the block diagram for programming a target device with the help of a PC and AVR board.

Fig. 2 shows the AVR microcontroller programmer circuit to burn the binary code into the target device. At the heart of the circuit is ATmega8515 (IC2). It features 8kB Flash memory with read-while-write capabilities, 512-byte EEPROM, 512-byte SRAM, 35 general-purpose input/output (I/O) lines, 32 general-purpose working registers, two flexible timers/counters with compare modes, internal and external interrupts, a serial programmable USART, a programmable watchdog timer with internal oscillator, and an SPI serial port.



Fig. 1: Block diagram for programming a target device with the help of a PC and AVR board

PARTS LIST

Semiconductors:
IC1, IC4 - 7805, 5V regulator
IC2, IC5 - ATmega8515 AVR microcontroller
IC3 - MAX232 RS-232 driver
D1-D4 - 1N4007 rectifier diode
LED1-LED9 - 5mm LED

Resistors (all ¼-watt, ±5% carbon)
R1, R11 - 10-kilo-ohm
R2-R10 - 330-ohm

Capacitors:
C1 - 1000µF, 25V electrolytic
C2, C15 - 0.1µF ceramic disk
C3, C14 - 4.7µF, 16V electrolytic
C4-C9 - 1µF, 16V electrolytic
C10-C13 - 22pF ceramic disk

Miscellaneous:
X1 - 230V AC primary to 9V, 500mA secondary transformer
X_{TAL1}, X_{TAL2} - 16MHz crystal
S1-S10 - Push-to-on tacktile switch
S11 - On/off switch
BATT. - 9V, battery
- 9-pin D-type female connector

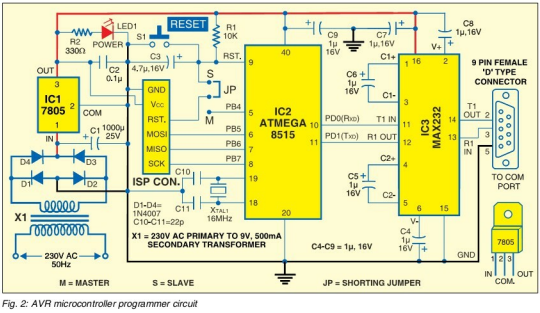


Fig. 2: AVR microcontroller programmer circuit

Switch S1 is used for manual reset. Port pins PB5 (MOSI), PB6 (MISO) and PB7 (SCK) are used for in-system programming (ISP). IC MAX232 (IC3) is used for serial communication. T_{XD} and R_{XD} pins of the microcontroller too are used for serial communication with the help of COM port.

In order to derive the power supply for the circuit, the 230V, 50Hz AC